

INTRODUCTION TO DATA MANAGEMENT

PROJECT REPORT

(Project Semester January-April 2025)



LOVELY
PROFESSIONAL
UNIVERSITY

GLOBAL HEALTH AND PROSPERITY INSIGHTS DASHBOARD

Submitted by

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Registration No: 12300581

Programme and Section: B.Tech. CSE KM005

Course Code: INT217

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Sweety Biju bearing Registration no. 12300581 has completed INT217 project titled, “**Global Health & Prosperity Insights Dashboard**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science & Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 13 April 2025

DECLARATION

I, Sweety Biju student of B. Tech. under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13 April 2025

Signature

A rectangular box containing a handwritten signature in black ink that reads "Sweety Biju".

Registration No. 12300581

Name of the student: Sweety Biju

TABLE OF CONTENTS

1. [ACKNOWLEDGEMENT](#)
2. [INTRODUCTION](#)
3. [SOURCE OF DATASET](#)
4. [EDA PROCESS](#)
5. [ANALYSIS ON DATASET](#)
6. [CONCLUSION](#)
7. [FUTURE SCOPE](#)
8. [REFERENCES](#)

ACKNOWLEDGEMENT

The completion of the “Global Health and Insights Dashboard” would not have been possible without the invaluable support, guidance, and encouragement from various individuals and institutions. I am deeply grateful to my college institution, **Lovely Professional University (LPU)**, for providing me with an exceptional academic environment, cutting-edge resources, and opportunities to explore and innovate in the fields of data science and socioeconomic analysis. The university’s dedication to fostering research, practical application, and analytical skills has been fundamental in shaping this project.

I extend my heartfelt thanks to my mentor, **Baljinder Kaur**, whose expertise, patience, and insightful feedback have been crucial in guiding me through the intricacies of data preprocessing, advanced visualization techniques, and comprehensive report development. Their unwavering support and constructive suggestions helped me navigate challenges, refine my methodologies, and ensure the project’s success.

I am also immensely thankful to my colleagues, for their collaboration, engaging brainstorming sessions, and technical assistance throughout this endeavor. Their diverse perspectives, willingness to share knowledge, and collaborative spirit enriched the project, making the process both intellectually stimulating and enjoyable.

This project stands as a testament to the collective effort and support I have received, and I am truly honored to have worked with such a dedicated team and institution. Thank you all for your contributions to this journey.

1. Introduction

The “Global Health & Prosperity Insights Dashboard” is a thorough and meticulously crafted tool that serves as a robust and dynamic platform for analyzing and visually presenting a wide array of socioeconomic data across a diverse range of nations worldwide. This project represents a significant effort to conduct a comprehensive and in-depth analysis of the dataset, with a deliberate and strategic focus on several critically important metrics that collectively paint a detailed picture of global conditions. These metrics include life expectancy, population distribution, infant mortality rates, HIV prevalence among adults, GDP per capita, and fertility rates. Each of these indicators has been thoughtfully selected to reflect both the economic strength and the overall health status of a variety of populations across different regions. By encompassing such a broad spectrum of data, the dashboard offers a multi-perspective view of the global scenario, enabling a deeper understanding of the interconnected factors that shape health outcomes and economic prosperity on an international scale.

The primary ambition of this project is to establish a strong and reliable foundation for evidence-based decision-making by identifying significant trends and illuminating the underlying patterns embedded within the data. These patterns and trends are intended to provide insightful and actionable information regarding worldwide health inequalities and economic success, shedding light on disparities that might otherwise go unnoticed. Through the meticulous identification of key problem areas—such as regions with persistently high infant mortality or low GDP per capita—alongside opportunities for development, this analysis seeks to equip researchers, policymakers, and other key stakeholders with the precise knowledge and evidence they need to design and implement focused plans and targeted interventions. The ultimate goal is to address pressing global challenges effectively and foster sustainable improvements in health and economic conditions.

This research is grounded in a custom-built dashboard that I have personally developed, which skillfully integrates a multitude of data points sourced from diverse origins and spanning various time periods. This integration is achieved through a unified and intuitive user interface that enhances accessibility and usability for all stakeholders. The dashboard’s design allows users to explore intricate relationships between different indicators—such as the correlation between GDP per capita and life expectancy—while deriving meaningful conclusions from its all-encompassing picture of the world. By providing such a holistic and interactive tool, the dashboard holds immense potential to encourage positive change, not only at a local level by addressing specific regional needs but also on a global scale by promoting equitable policies and practices that can benefit populations everywhere. This ambitious project thus stands as a testament to the power of data-driven analysis in driving transformative impact across the globe.

2. Source of Dataset

The primary dataset for this analysis was sourced from the UNESCO Institute for Statistics Data Browser, accessible at https://databrowser.uis.unesco.org/browser/DEMOGRAPHIC_SOCIOECONOMIC/UIS-DemSocEco, which provides a comprehensive collection of some demographic and socioeconomic indicators. Additionally, secondary datasets were obtained from the World Bank's open data portal at <https://data.worldbank.org/>, offering detailed economic and development metrics, and from https://en.wikipedia.org/wiki/List_of_ISO_3166_country_codes, which supplied country code mappings. The data encompasses various metrics collected over multiple years, allowing for a longitudinal analysis of trends. I obtained the datasets in CSV format, which were then imported into the dashboard for further processing and visualization.

3. Dataset Preprocessing

To ensure accuracy, consistency, and suitability for the necessary visualisations and insights in the "Global Health & Prosperity Insights Dashboard," the dataset was subjected to a number of preprocessing stages prior to the detailed analysis. To transformed the raw data into a solid and accurate foundation for further analysis, several procedures were underwent. The steps are described as follows:

- Initial Data Cleaning:
 - Removed unwanted sheets that were irrelevant to the analysis, such as “footnotes”, “disaggregations”, and “glossary terms”, which could introduce confusion.
 - Deleted blank columns, like "qualifier" and “magnitude”, to remove unnecessary variables that did not contribute to the analysis.

	A	B	C	D	E	F	G	H	I
1	indicator	geoUnit	year	value	qualifier	magnitude			
2	200101	ABW	2000	87.822					
3	200101	ABW	2001	89.701					
4	200101	ABW	2002	90.909					
5	200101	ABW	2003	91.848					
6	200101	ABW	2004	92.772					
7	200101	ABW	2005	93.653					
8	200101	ABW	2006	94.807					
9	200101	ABW	2007	95.978					
10	200101	ABW	2008	97.196					
11	200101	ABW	2009	98.413					
12	200101	ABW	2010	99.638					
13	200101	ABW	2011	100.589					
14	200101	ABW	2012	101.373					
15	200101	ABW	2013	102.154					
16	200101	ABW	2014	102.961					
17	200101	ABW	2015	103.784					
18	200101	ABW	2016	104.585					
19	200101	ABW	2017	105.368					
20	200101	ABW	2018	106.093					
21	200101	ABW	2019	106.771					
22	200101	ABW	2020	107.381					
23	200101	ABW	2021	107.42					
24	200101	ABW	2022	107.696					
25	200101	ABW	2023	107.868					
26	200101	AFG	2000	20251.96					
27	200101	AFG	2001	20008.7					
28	200101	AFG	2002	20559.92					
29	200101	AFG	2003	22196.32					
30	200101	AFG	2004	23269.78					
31	200101	AFG	2005	23851.53					

< >
Data
Footnotes
Indicators
Disaggregations
Glossary Terms

Fig. 3.1 Removing unwanted sheets and columns

- Indicator Mapping and Restructuring:
 - The indicator codes were mapped to their corresponding actual names, replacing cryptic abbreviations (not understandable by an average person) with descriptive labels (e.g., infant mortality rate or GDP per capita) to enhance user understanding.
 - The dataset was reorganized according to my preference. It was restructured so that indicators were presented as columns rather than rows, helping in easier access and comparison across the different indicators.

A	B
Country	geoUnit
Afghanistan	=IFERROR(VLOOKUP([@Country],Table2,2,FALSE),"UNKNOWN")
Afghanistan	=IFERROR(VLOOKUP([@Country],Table2,2,FALSE),"UNKNOWN")
Afghanistan	=IFERROR(VLOOKUP([@Country],Table2,2,FALSE),"UNKNOWN")
Afghanistan	=IFERROR(VLOOKUP([@Country],Table2,2,FALSE),"UNKNOWN")
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Albania	=IFERROR(VLOOKUP([@Country],Table2,2,FALSE),"UNKNOWN")
Albania	=IFERROR(VLOOKUP([@Country],Table2,2,FALSE),"UNKNOWN")
Albania	=IFERROR(VLOOKUP([@Country],Table2,2,FALSE),"UNKNOWN")

Fig 3.2 Indicator Mapping

- Data Filtering:
 - The data for the year 2023 was rather sparse and incomplete. Hence there was deliberate decision to delete these records to ensure the overall integrity of the analysis, focusing instead on the more complete data from 2000 to 2022.
- Geographical Standardization:
 - The country codes in the "geoUnit" column were based on the ISO 3166 standard. These were mapped to their corresponding understandable country names by utilising Wikipedia's country-code mapping data. The mapping was achieved using the VLOOKUP excel formula for efficient matching.
 - Further standardized these country names by converting less commonly used terms—such as "People's Republic of Korea" to "North Korea"—using the FIND AND REPLACE Option, ensuring consistency and familiarity to reduce potential misinterpretations.

	A
1	Country
618	Korea (the Democratic People's Republic of) [t]
619	Korea (the Democratic People's Republic of) [t]
620	Korea (the Democratic People's Republic of) [t]
621	Korea (the Democratic People's Republic of) [t]
622	Korea (the Democratic People's Republic of) [t]
623	Korea (the Democratic People's Republic of) [t]

Fig 3.3(i)

A	B
Country	GeoUnit
North Korea	PRK
North Korea	PRK
North Korea	PRK
North Korea	PRK
North Korea	PRK
North Korea	PRK

Fig 3.3(ii)

	Country	Column1	year	Fertility ra
4898	Tanzania, the United Republic of		2000	
4899	Tanzania, the United Republic of		2001	
4900	Tanzania, the United Republic of		2002	
4901	Tanzania, the United Republic of		2003	
4902	Tanzania, the United Republic of		2004	
4903	Tanzania, the United Republic of		2005	
4904	Tanzania, the United Republic of		2006	
4905	Tanzania, the United Republic of		2007	
4906	Tanzania, the United Republic of		2008	
4907	Tanzania, the United Republic of		2009	
4908	Tanzania, the United Republic of		2010	
4909	Tanzania, the United Republic of		2011	
4910	Tanzania, the United Republic of		2012	
4911	Tanzania, the United Republic of		2013	
4912	Tanzania, the United Republic of		2014	
4913	Tanzania, the United Republic of		2015	

Fig 3.3(iii) Geographical Standardization

- Data Enrichment:
 - Added GDP per capita data by creating a unique key that combined country and year information, then used the VLOOKUP formula to match this key with corresponding GDP per capita values from another sheet, enhancing the economic dimension of the analysis.

per 1,000	Population growth (annual %)	Prevalence of HIV, total (% of population ages 15)	Total population (thousands)
	3.46446	1.7	30763.554
	3.39528	1.6	31830.756
	3.26835	1.6	32920.509
	3.16603	1.5	33981.756
	3.09675	1.5	35083.102
	3.101	0.05	11.002
	$=(((\text{Total population (thousands)})-M117)/M117)*100$	0.05	11.351
	$=(((\text{Total population (thousands)})-M118)/M118)*100$	0.05	11.592
	$=(((\text{Total population (thousands)})-M119)/M119)*100$	0.05	11.781
	$=(((\text{Total population (thousands)})-M120)/M120)*100$	0.05	11.968
	$=(((\text{Total population (thousands)})-M121)/M121)*100$	0.05	12.167
	$=(((\text{Total population (thousands)})-M122)/M122)*100$	0.05	12.366
	$=(((\text{Total population (thousands)})-M123)/M123)*100$	0.05	12.573
	$=(((\text{Total population (thousands)})-M124)/M124)*100$	0.05	12.779
	$=(((\text{Total population (thousands)})-M125)/M125)*100$	0.05	13
	$=(((\text{Total population (thousands)})-M126)/M126)*100$	0.05	13.227
	$=(((\text{Total population (thousands)})-M127)/M127)*100$	0.05	13.462
	$=(((\text{Total population (thousands)})-M128)/M128)*100$	0.05	13.677
	$=(((\text{Total population (thousands)})-M129)/M129)*100$	0.05	13.081
	$=(((\text{Total population (thousands)})-M130)/M130)*100$	0.05	13.423

Fig 3.5(ii) Handled Missing Values

- Data Formatting:
 - Number-formatted all numerical columns, applying appropriate decimal places and units to enhance clarity.

Number		Conditional	Format as	Cell
Number		Formatting	Table	Style
a Bar		J		
-49)	Total population (thousands)			
0.20	23586.69			
0.20	23745.13			
0.20	23885.59			
0.20	24002.36			
0.20	24178.40			

Fig 3.6 Number Formatting

- Regional Mapping:
 - Mapped each country to its widely recognized continent—such as Africa, Asia, Australia, Europe, North America, and South America—using a predefined mapping table fetched from trusted sites, enabling grouped regional analysis to complement country-specific data.

A	B
Country	CONTINENT
Laos	Asia
Sint Maarten	North America
Afghanistan	Asia
Albania	Europe
Algeria	Africa
American Samoa	Africa

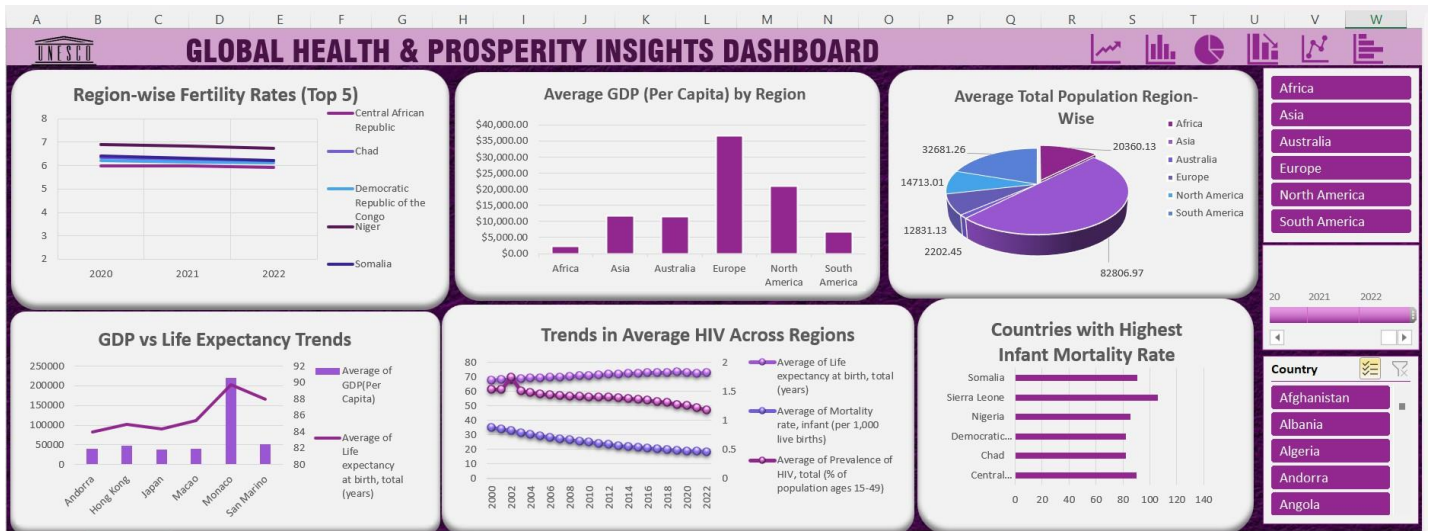
Fig 3.7 Country-Continent Mapping

Final Dataset Structure:

The preprocessed dataset includes the following columns: "Country (showing the standardized country names)", "geoUnit (i.e. country codes adhering to the ISO 3166 standards)", "Year," "GDP (Per Capita)", "Fertility rate, total (births per woman)", "Life expectancy at birth, total (years)", "Mortality rate, infant (per 1,000 live births)", "Population growth (annual %)", "prevalence of HIV, total (% of population)", "Total population (thousands)", and "Continents". This structured and enriched dataset provided a solid foundation for the dashboard, ensuring that the subsequent analysis of trends and disparities across health and economic indicators was both accurate and meaningful.

A	B	C	D	E	F	G	H	I	J	K
Country	GeoUnit	Year	GDP(Per Capita)	Fertility rate, total (births per woman)	Life expectancy at birth, total (years)	Mortality rate, infant (per 1,000 live births)	Population growth (annual %)	Prevalence of HIV, total (% of population ages 15-49)	Total population (thousands)	Continent
India	IND	01-01-2000	442.03533	3.35	62.67	66.40	1.82	0.6	378.254	Asia
India	IND	01-01-2001	449.91031	3.30	63.09	64.50	1.81	0.5	390.254	Asia
India	IND	01-01-2002	468.84551	3.22	63.62	62.60	1.78	0.5	1087954.254	Asia
India	IND	01-01-2003	543.84505	3.12	64.09	60.60	1.72	0.5	1107246.507	Asia
India	IND	01-01-2004	624.1088	3.05	64.52	58.70	1.67	0.5	1126359.505	Asia
India	IND	01-01-2005	710.51122	2.96	65.00	56.80	1.60	0.4	1145623.522	Asia
India	IND	01-01-2006	802.01374	2.86	65.41	54.80	1.52	0.4	1163729.122	Asia
India	IND	01-01-2007	1022.73247	2.78	65.79	52.80	1.47	0.4	1182028.657	Asia
India	IND	01-01-2008	993.50941	2.72	66.15	50.80	1.42	0.4	1199323.384	Asia
India	IND	01-01-2009	1096.63614	2.67	66.51	48.70	1.39	0.3	1216538.543	Asia
India	IND	01-01-2010	1350.69447	2.60	66.91	46.70	1.38	0.3	1234510.963	Asia
India	IND	01-01-2011	1449.6033	2.54	67.36	44.60	1.36	0.3	1252452.165	Asia
India	IND	01-01-2012	1434.01799	2.47	67.89	42.50	1.33	0.3	1269997.744	Asia
India	IND	01-01-2013	1438.05701	2.41	68.46	40.50	1.30	0.3	1287351.261	Asia
India	IND	01-01-2014	1559.86378	2.31	69.07	38.50	1.24	0.3	1304307.76	Asia
India	IND	01-01-2015	1590.17433	2.29	69.64	36.50	1.19	0.3	1320246.623	Asia
India	IND	01-01-2016	1714.27954	2.27	70.12	34.70	1.19	0.2	1335802.372	Asia
India	IND	01-01-2017	1957.96981	2.20	70.47	32.90	1.16	0.2	1352086.22	Asia
India	IND	01-01-2018	1974.37773	2.18	70.71	31.30	1.09	0.2	1367228.58	Asia
India	IND	01-01-2019	2050.1638	2.11	70.91	29.70	1.03	0.2	1382089.548	Asia
India	IND	01-01-2020	1915.55159	2.05	70.15	28.10	0.96	0.2	1395971.076	Asia
India	IND	01-01-2021	2250.17902	2.03	67.24	26.80	0.80	0.2	1409264.315	Asia
India	IND	01-01-2022	2366.30961	2.01	67.74	25.50	0.68	0.2	1419143.477	Asia

Fig 3.8 Final Dataset



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4. Analysis on Dataset (for each objective)

Objective 1: To compare the economic prosperity (measured by GDP per capita) with health outcomes (measured by Life Expectancy) across continents, enabling users to interactively explore country-specific contributions to these continental averages, thereby identifying economic-health disparities and potential areas for targeted development interventions.

i. General Description:

- This objective aims to evaluate the relationship between economic prosperity, measured by GDP per capita, and health outcomes, represented by life expectancy, across continents such as Africa, Asia, Australia, Europe, North America, and South America. The goal is to understand how economic conditions influence health outcomes and to identify regions with significant disparities.
- It utilizes the preprocessed GDP per capita and life expectancy data to categorize continental performance and assess country-specific contributions to these averages.
- The primary aim is to highlight continents and countries where low GDP per capita correlates with reduced life expectancy, enabling policymakers to target development interventions effectively.

- By providing a comparative analysis across continents, this objective facilitates the identification of trends and imbalances, supporting informed decision-making to enhance global health equity.

ii. Specific Requirements:

- Pivot Table Usage:
 - An Excel pivot table named "GDP_LifeExpectancy" was used to consolidate and compare average GDP per capita and life expectancy by continent and was used as the source table for the "GDP vs Life Expectancy Trends" visualization.
- Rows Included: The COUNTRY column was used as the row label.
- Values Included:
 - Average of GDP Per Capita: Mean of the "GDP Per Capita" column by country, presenting an idea of economic prosperity.
 - Average of Life Expectancy: Average of the "Life Expectancy" column by country, indicating health outcomes.
- Filters Added:
 - A value filter was added on "COUNTRY" to focus on top 6 countries by Average GDP per Capita enhancing and maintaining visual clarity.

iii. Analysis Results:

- The "GDP vs Life Expectancy Trends" combo chart reveals a strong positive correlation between GDP per capita and life expectancy across continents. For instance, North American countries like Bermuda, with average GDP per capita around \$130000 show life expectancies exceeding 80 years. In contrast, African countries like Equatorial Guinea, with a GDP per capita of approximately \$8,000, has a life expectancy around 60 years.
- Country-specific contributions, such as Japan and Switzerland (high GDP and life expectancy above 85 years), contrast with countries like Chad and Somalia (low GDP and life expectancy below 65 years), highlighting significant disparities.
- The analysis suggests that economic prosperity is a critical determinant of health outcomes, with wealthier regions investing more in healthcare infrastructure, while poorer regions face challenges in accessing such resources.

- These findings indicate a need for targeted economic development and health interventions in low-GDP regions like Africa to bridge the gap.

iv. Visualization:

- Visualization Used: Combo Chart
- The "GDP vs Life Expectancy Trends" chart employs a Combo chart, with bars representing average GDP per capita (purple) and a line representing life expectancy (pink) side by side for top 6 countries by average GDP in the continent (selectable through slicers).
- The bars effectively depict the variation in economic prosperity and health outcomes, with taller bars in North America and Europe contrasting with shorter bars in Africa.
- Reason for Choice: The chart was chosen to accommodate the different scales of GDP per capita (in dollars) and life expectancy (in years), allowing clear comparison without distortion. The purple and pink color scheme enhances readability and aligns with the dashboard's aesthetic, enabling stakeholders to easily interpret disparities and make informed decisions.

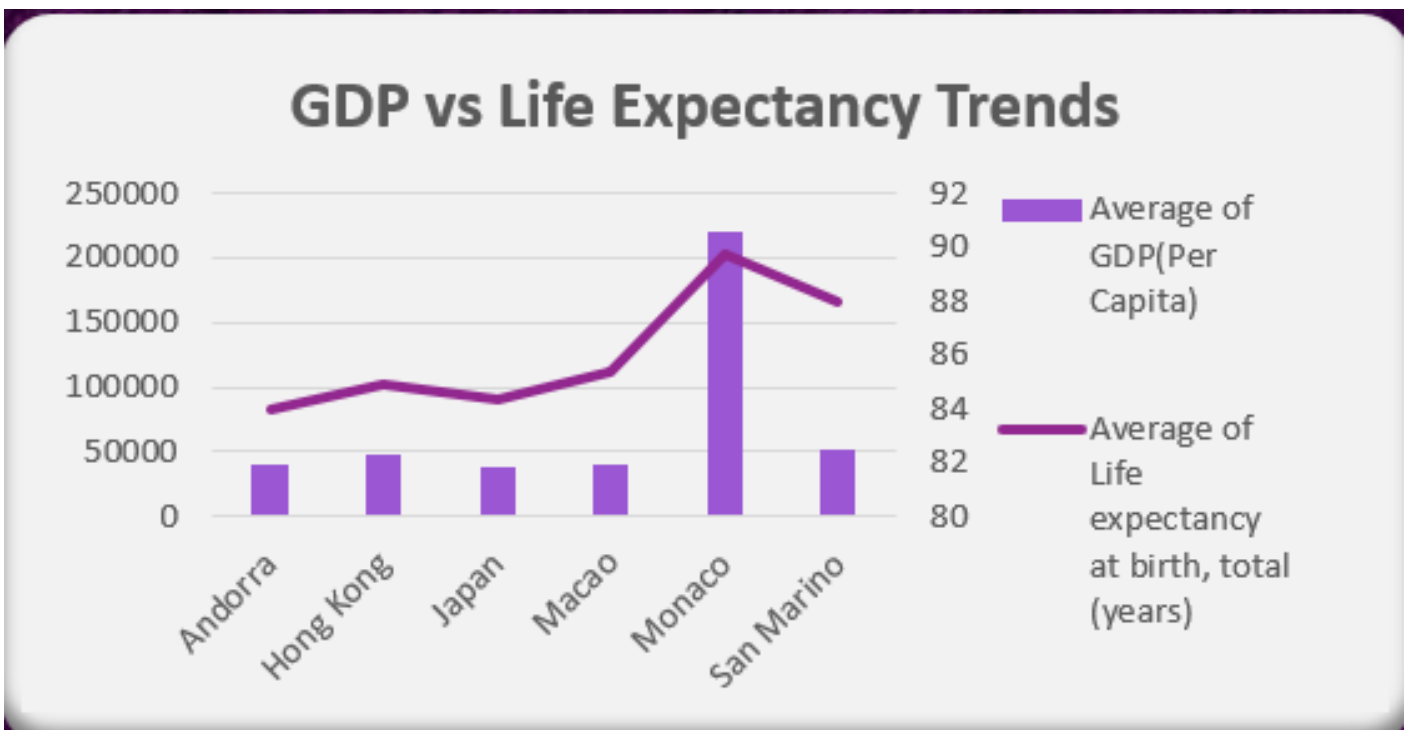


Fig. 4.1 GDP per Capita vs Life Expectancy

Objective 2: To analyze the long-term trends in HIV prevalence and its impact on Life Expectancy and Infant Mortality rates across regions over the years, enabling users to identify patterns of health improvement or deterioration and inform targeted public health strategies to address HIV-related challenges.

i. General Description:

- This objective seeks to examine the long-term trends in HIV prevalence across regions (Africa, Asia, Australia, Europe, North America, South America) from 2000 to 2022 and assess its impact on life expectancy and infant mortality rates.
- It utilizes the preprocessed HIV prevalence data, alongside life expectancy and infant mortality rate data, to identify correlations and regional variations.
- The primary aim is to detect patterns of health improvement or deterioration, particularly in high-prevalence regions like Africa, to guide public health strategies and resource allocation.
- By analyzing these trends, this objective supports the development of targeted interventions to mitigate HIV's impact on population health.

ii. Specific Requirements:

- Pivot Table Usage:
 - An Excel pivot table named "HIV_Trends" was used to compare average HIV prevalence, life expectancy, and infant mortality rates by region and was used as the source pivot table for the "Trends in Average HIV Across Regions" visualization.
- Rows Included:
 - The YEAR column was employed as the row label, to show temporal changes.
- Values Included:
 - Average of HIV Prevalence: Calculated as the mean of the "HIV Prevalence" column.
 - Average of Life Expectancy: Computed as the average of the "Life Expectancy" column.
 - Average of Infant Mortality Rate: Calculated as the average of the "Infant Mortality Rate" column.

iii. Analysis Results:

- The "Trends in Average HIV Across Regions" line graph shows a declining trend in HIV prevalence across all regions since 2000, with Africa historically having the highest rates (peaking

above 6% in 2000, now below 3.5%). This decline correlates with improvements in life expectancy (e.g., from 55 to 60 years in Africa).

- Infant mortality rates remain high in regions with past high HIV prevalence (above 60 per 1,000 live births), suggesting a lagged impact of HIV on child health.
- The analysis indicates that public health interventions, such as antiretroviral therapy, have been effective, but challenges persist in regions with residual high infant mortality, necessitating continued focus.

iv. Visualization:

- Visualization Used: Line Graph(with Markers)
- The "Trends in Average HIV Across Regions" chart uses a line graph, with lines representing HIV prevalence , life expectancy and infant mortality rate over time for each region.
- The lines effectively track trends, with declining HIV prevalence and rising life expectancy contrasting with persistent high infant mortality in some regions.
- Reason for Choice: The line graph was chosen to depict temporal trends clearly, accommodating the continuous nature of the data. The color scheme enhances trend visibility, aiding stakeholders in identifying improvement areas.

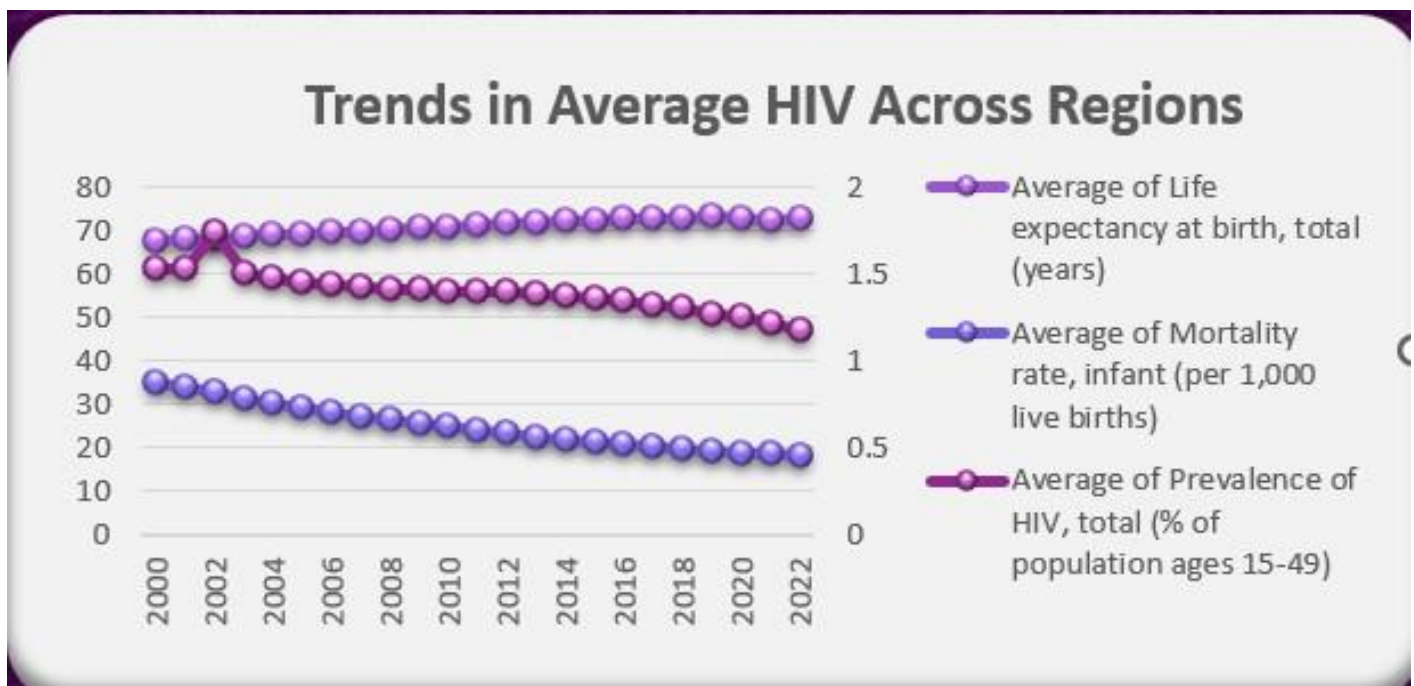


Fig. 4.2 Trends in Average HIV Across Regions

Objective 3: To identify the countries with the highest infant mortality rates, enabling users to prioritize regions for targeted health interventions and resource allocation, while providing a comparative overview to assess the severity of infant health challenges across these high-risk nations.

i. General Description:

- This objective aims to identify the top countries with the highest infant mortality rates across regions, such as Somalia, Sierra Leone, Afghanistan, Chad, and Nigeria, to prioritize regions for health interventions.
- It uses the preprocessed infant mortality rate data to rank countries and assess the severity of infant health challenges.
- The primary aim is to highlight regions requiring urgent resource allocation and policy focus to reduce infant mortality.
- By providing a comparative overview, this objective supports targeted strategies to address disparities in infant health outcomes.

ii. Specific Requirements:

- Pivot Table Usage:
 - An Excel pivot table named "InfantMortality_TopCountries" was used to rank and compare infant mortality rates by country and was used as the source for the "Countries with Highest Infant Mortality Rate" visualization.
- Rows Included:
 - The COUNTRY column was employed as the row label, aggregating data for countries with the highest rates.
- Values Included:
 - Average of Infant Mortality Rate: Calculated as the mean of the "Infant Mortality Rate" column by country.
- Filters Added:
 - A value filter on "COUNTRY" was included to focus on the top 6 countries.
- Slicers allowing Continent wise Highest Infant Mortality Rate Countries to be shown is added to increase interactivity.

iii. Analysis Results:

- The "Countries with Highest Infant Mortality Rate" horizontal bar chart identifies Somalia, Sierra Leone, Chad, and Nigeria as the top countries across continents, with rates exceeding 80-100 per 1,000 live births, far above the global average.
- These countries, predominantly in Africa and parts of Asia, reflect challenges in healthcare access and infrastructure, likely exacerbated by economic and conflict-related factors.
- The analysis suggests a need for targeted interventions, such as improved maternal and neonatal care, in these high-risk nations.

iv. Visualization:

- Visualization Used: Horizontal Bar Chart
- The "Countries with Highest Infant Mortality Rate" chart uses a horizontal bar chart, with bars representing infant mortality rates (purple) for each country.
- The bars help clearly visualize the mortality rates.
- Reason for Choice: The horizontal bar chart was chosen for its ability to rank and compare countries easily, with the purple color enhancing readability for stakeholders.

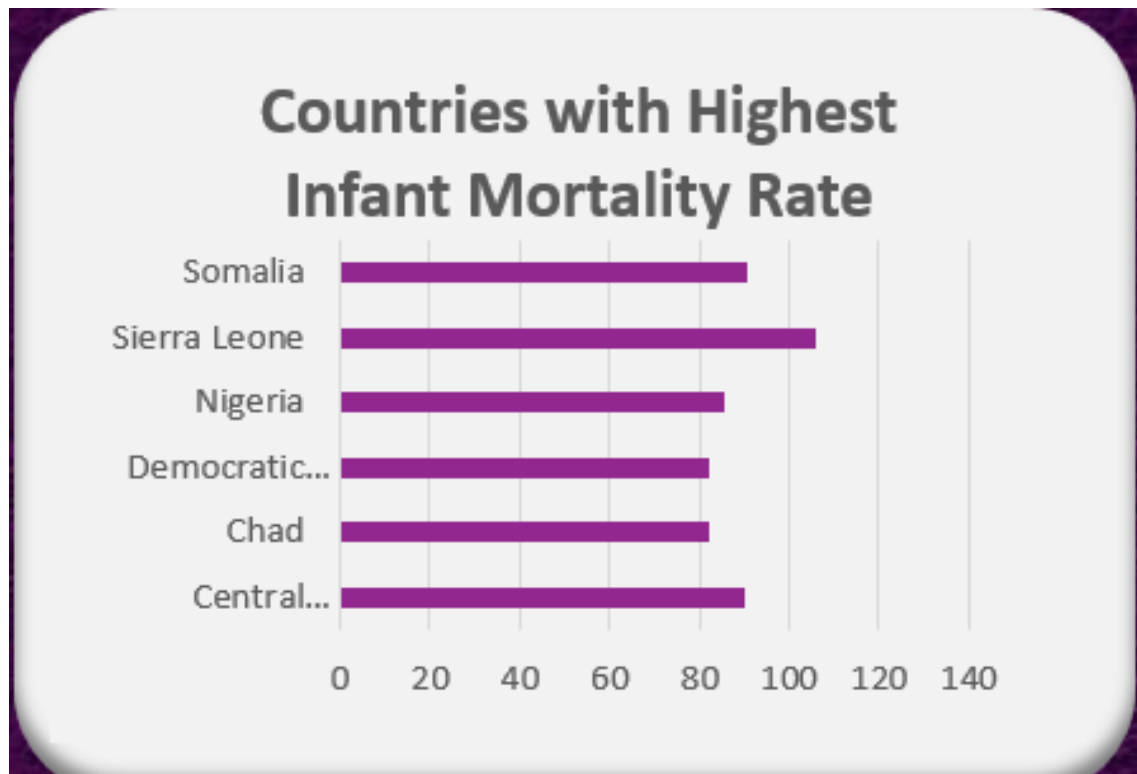


Fig 4.3 Countries with Highest IMR

Objective 4: To assess the regional distribution of average population sizes, enabling users to understand the demographic weight of each region and its potential influence on health and economic indicators, thereby guiding resource allocation and policy prioritization for development initiatives.

i. General Description:

- This objective aims to assess the average population sizes across regions (Africa, Asia, Australia, Europe, North America, South America) to understand their demographic weight and influence on health and economic indicators.
- It utilizes the preprocessed population data to categorize regional distributions and evaluate their implications.
- The primary aim is to guide resource allocation and policy prioritization based on population size and its associated demands.
- By providing a demographic overview, this objective supports balanced development strategies across regions.

ii. Specific Requirements:

- Pivot Table Usage:
 - An Excel pivot table named "Population_Distribution" was utilized to help compare average population sizes by region and was used as the source table for the "Average Total Population by Region" visualization.
- Rows Included:
 - The "CONTINENT" column was employed as the row label, aggregating data for regions.
- Values Included:
 - Average of Total Population: Calculated as the mean of the "Total Population" column by region.

iii. Analysis Results:

- The "Average Total Population by Region" pie chart shows Asia with the largest share (4,731.13 million), followed by Africa (3,268.26 million), Europe (1,473.13 million), North America (828.97 million), South America (220.45 million), and Australia.

- This distribution suggests that Asia and Africa's large populations exert significant pressure on health and economic resources, while smaller regions like Australia have less demographic influence.
- The analysis indicates a need for proportional resource allocation to address the needs of densely populated regions.

iv. Visualization:

- Visualization Used: Pie Chart
- The "Average Total Population by Region" chart uses a pie chart, with segments representing population shares (purple shades) for each region.
- The pie chart effectively highlights the dominance of Asia and Africa.
- Reason for Choice: The pie chart was chosen to depict proportional distribution clearly, with the purple palette enhancing visual appeal for stakeholders.

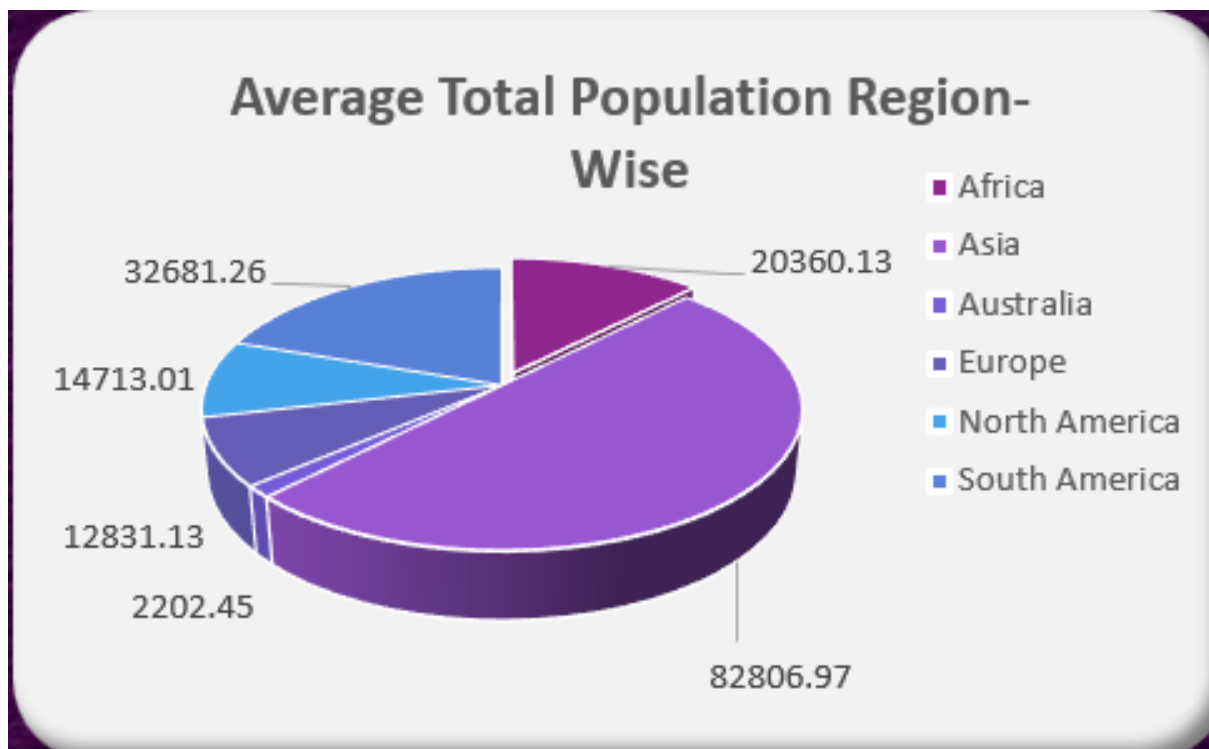


Fig. 4.4 Average Total Population Region-wise

Objective 5: To analyze the trends in fertility rates over time across countries with the highest rates, enabling users to identify patterns of demographic change and assess their implications for population growth, health resource planning, and economic development strategies across regions.

i. General Description:

- This objective aims to analyze the trends in fertility rates over time (2000–2022) for countries with the highest rates, such as Chad, Democratic Republic of the Congo, Mali, Niger, and Somalia.
- It utilizes the preprocessed fertility rate data to track changes and evaluate their impact on population growth and resource planning.
- The primary aim is to identify patterns of demographic change and inform health and economic development strategies.
- By focusing on high-fertility countries, this objective supports proactive policy measures to manage population dynamics.

ii. Specific Requirements:

- Pivot Table Usage:
 - An Excel pivot table named "Fertility_Trends" was utilized to compare fertility rates by country and was used as the source for the "Region-wise Fertility Rates (Top 5)" visualization.
- Rows Included:
 - The “YEAR” column was employed as the row label, depicting temporal changes.
- Columns Included:
 - The “COUNTRY” column was employed as the column label, aggregating data for the top 5-6 countries.
- Values Included:
 - Average of Fertility Rate: Calculated as the mean of the “Fertility Rate” column by country and year.
- Filters Added:
 - A filter on “COUNTRY” was included to focus on the top five countries.

iii. Analysis Results:

- The "Region-wise Fertility Rates (Top 5)" line graph shows a declining trend in fertility rates for Chad, Democratic Republic of the Congo, Mali, Niger, and Somalia in African continent, from around 7 births per woman in 2000 to below 6 in 2022.
- This decline suggests shifts due to increased education and healthcare access, though rates remain high compared to global averages.
- The analysis indicates implications for population growth and resource planning, requiring sustained health and economic investments.

iv. Visualization:

- Visualization Used: Line Graph
- The "Region-wise Fertility Rates (Top 5)" chart uses a line graph, with lines representing fertility rates (purple) for each country over time.
- The lines effectively track the downward trend across countries.
- Reason for Choice: The line graph was chosen to depict temporal trends clearly, with the purple color enhancing readability for stakeholders.

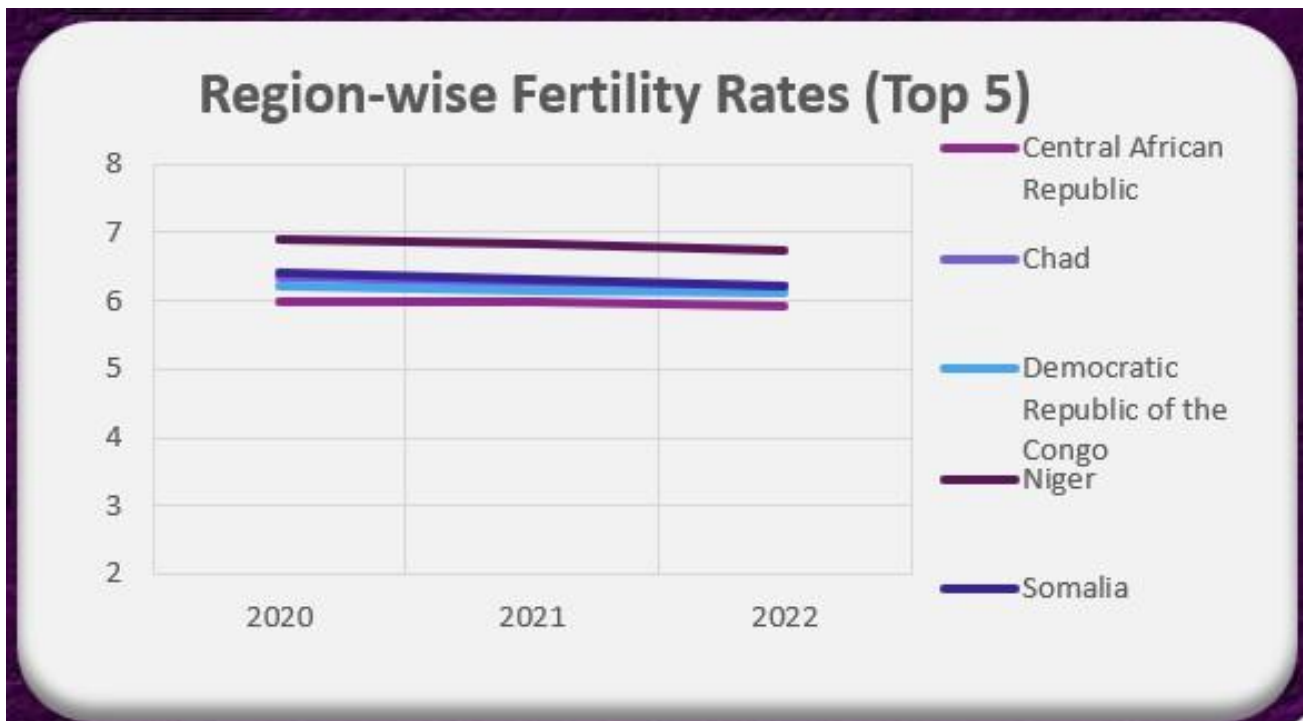


Fig 4.5 Region-wise Fertility Rates(Top 5 Countries)

Objective 6: To evaluate the regional disparities in average GDP per capita, enabling users to identify economic inequalities and their potential correlation with health indicators, thereby guiding economic development and health policy initiatives.

i. General Description:

- This objective aims to evaluate regional disparities in average GDP per capita across Africa, Asia, Australia, Europe, North America, and South America to identify economic inequalities and their potential correlation with health indicators.
- It utilizes the preprocessed GDP per capita data to categorize regional economic performance and assess its impact on health outcomes.
- The primary aim is to guide economic development and health policy initiatives by highlighting regions with significant economic challenges.
- By providing a regional economic overview, this objective supports strategies to address inequalities and improve health equity.

ii. Specific Requirements:

- Pivot Table Usage:
 - An Excel pivot table named "GDP_PerCapita_Region" was utilized to consolidate and compare average GDP per capita by region and was used as the source for the "Average GDP Per Capita by Region" visualization.
- Rows Included:
 - The “CONTINENT” column was employed as the row label, aggregating data for regions.
- Values Included:
 - Average of GDP Per Capita: Calculated as the mean of the “GDP Per Capita” column by region.
- Slicers and timelines deployed for additional interactivity.

iii. Analysis Results:

- The "Average GDP Per Capita by Region" bar chart shows North America and Europe having the highest GDP per capita, while Africa has the lowest (around \$10,000).
- The slight variances in the data probably determine fewer records for some continents.
- This disparity correlates with health indicators, such as higher life expectancy in North America and Europe versus lower rates in Africa.
- The analysis suggests that economic inequalities drive health disparities, necessitating targeted economic and health policies in regions like Africa.

iv. Visualization:

- Visualization Used: Column Chart
- The "Average GDP Per Capita by Region" chart uses a bar chart, with bars representing GDP per capita (purple) for each region.
- The bars clearly depict the economic disparity, with North America's bar being the tallest.
- Reason for Choice: The column chart was chosen for its ability to compare regional economic performance clearly, with the purple color enhancing visibility for stakeholders.

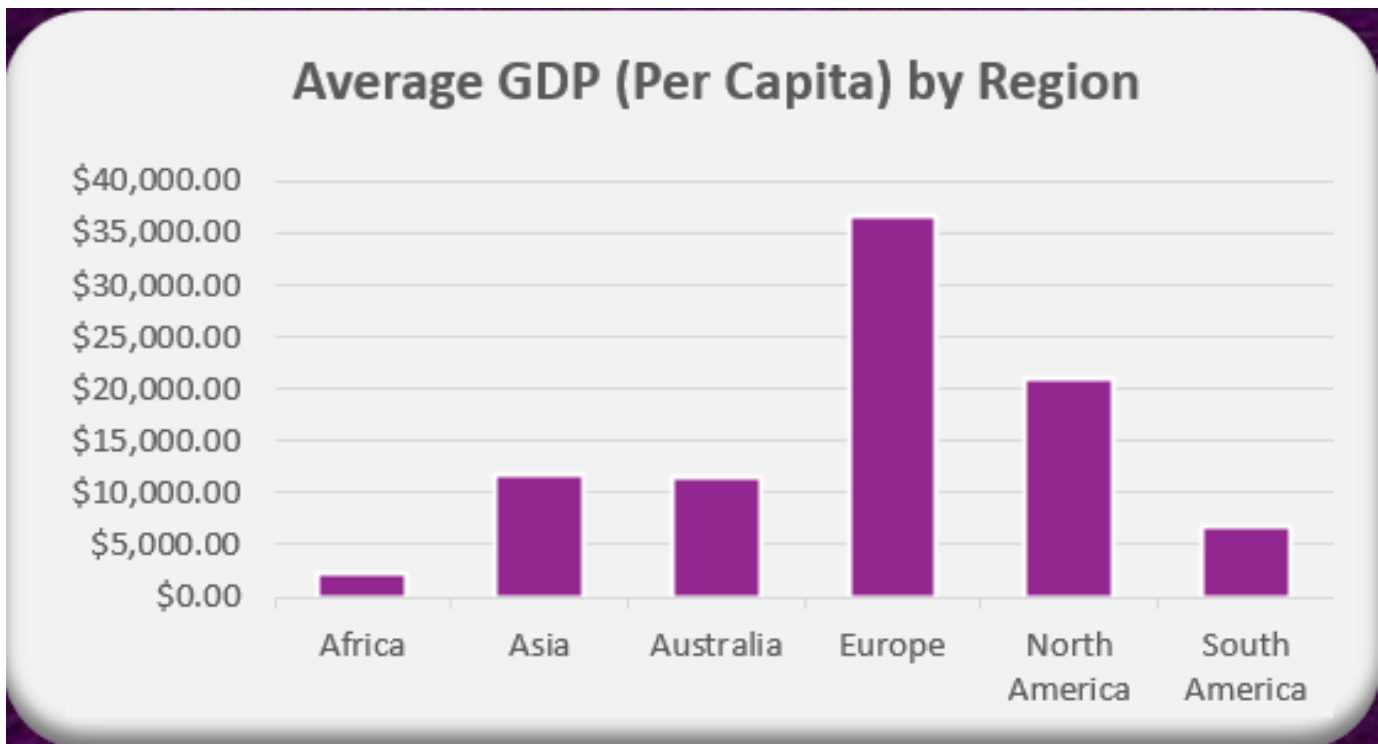


Fig 4.6 Average GDP Per Capita by Region

5. Conclusion

The analysis conducted through the 'Global Health & Prosperity Insights Dashboard' has revealed significant and notable differences in both economic and health indicators among various nations and regions across the globe, providing a detailed and revealing snapshot of global disparities. Based on a thorough and meticulous examination of the data, it has become strikingly clear that many African nations, along with certain countries in Asia, are grappling with persistently high fertility rates and alarmingly high infant mortality rates. These challenging health metrics appear to be directly and inextricably linked to their comparatively low GDP per capita, suggesting that economic constraints play a critical role in shaping these outcomes. In stark contrast, regions such as North America and Europe are shown to benefit from greater economic success, as evidenced by their substantially higher GDP per capita figures, which are closely associated with longer life expectancies. These extended life expectancies serve as a strong indicator of overall better health outcomes, reflecting the advantages of robust healthcare systems and economic stability in these areas.

Furthermore, the analysis has uncovered encouraging and positive downward trends in both HIV prevalence and fertility rates over the observed time period, which stand as compelling evidence of the significant advancements and progress achieved through international health programs and targeted interventions. These improvements suggest that global efforts to combat diseases and manage population growth are yielding tangible results, offering hope for continued enhancement in public health. However, despite these encouraging advancements, a multitude of persistent problems remain, particularly in regions characterized by limited infrastructure and scarce resources. These ongoing challenges underscore the complex and multifaceted relationship between economic circumstances and health consequences, highlighting the urgent and pressing need for focused and strategic interventions to address these disparities effectively.

The findings from this analysis emphasize the intricate interplay between economic conditions and health outcomes, revealing how economic prosperity—or the lack thereof—can profoundly influence the well-being of populations. To navigate and address these complexities, the dashboard I have personally created emerges as a particularly useful and indispensable tool. It provides an organized, transparent, and user-friendly framework that enables a deep comprehension of these intricate interactions, making the data accessible and actionable for all stakeholders. By offering critical insights derived from the analysis, the dashboard serves as

a vital resource for policymakers, researchers, academics, and other key decision-makers. These insights have the potential to significantly influence and direct future policy decisions, fostering a more equitable and inclusive approach to global health and economic development. This tool not only illuminates the current state of affairs but also paves the way for informed strategies that can help bridge the gap between resource-rich and resource-poor regions, ultimately contributing to a more balanced and prosperous world.

6. Future Scope

The analysis conducted through the ****Global Health & Prosperity Insights Dashboard**** has revealed significant and notable differences in both economic and health indicators among a diverse array of nations and regions spanning the globe, delivering a detailed, revealing, and comprehensive snapshot of the prevailing global disparities. Through a thorough and meticulous examination of the extensive dataset, it has become strikingly clear that many nations across Africa, alongside certain countries in Asia, are currently grappling with persistently high fertility rates and alarmingly high infant mortality rates. These challenging and deeply concerning health metrics appear to be directly and inextricably linked to their comparatively low GDP per capita, strongly suggesting that economic constraints and limited financial resources play a critical and foundational role in shaping these adverse outcomes. The data indicates that the lack of economic prosperity in these regions restricts access to essential healthcare services, education, and infrastructure, perpetuating a cycle of poor health and economic stagnation. In stark contrast, regions such as North America and Europe are shown to benefit from greater economic success, as evidenced by their substantially higher GDP per capita figures, which are closely and consistently associated with longer life expectancies. These extended life expectancies, often exceeding 80 years in these areas, serve as a robust and compelling indicator of overall better health outcomes, reflecting the significant advantages provided by robust healthcare systems, advanced medical technologies, and the economic stability that enables sustained investment in public welfare.

Furthermore, the analysis has uncovered encouraging and positive downward trends in both HIV prevalence and fertility rates over the observed time period from 2000 to 2022, which stand as compelling and heartening evidence of the significant advancements and substantial progress achieved through the dedicated efforts of international health programs and carefully targeted interventions. These improvements suggest that the global community's concerted efforts to combat infectious diseases like HIV and to manage population growth through education and healthcare access are yielding tangible and measurable results, offering a beacon of

hope for continued enhancement in public health on a worldwide scale. The decline in HIV prevalence, particularly in high-burden regions like Africa, and the gradual reduction in fertility rates in countries with historically high birth rates, point to the effectiveness of initiatives such as antiretroviral therapy distribution and family planning programs. However, despite these encouraging advancements, a multitude of persistent and deeply entrenched problems remain, particularly in regions characterized by limited infrastructure, scarce resources, and ongoing socio-political challenges. These ongoing issues underscore the complex, multifaceted, and often interdependent relationship between economic circumstances and health consequences, highlighting the urgent and pressing need for focused, strategic, and well-coordinated interventions to address these disparities effectively and sustainably.

The findings from this extensive analysis emphasize the intricate and dynamic interplay between economic conditions and health outcomes, revealing how the presence or absence of economic prosperity can profoundly and far-reachingly influence the well-being, quality of life, and longevity of populations across the globe. To navigate and address these complexities with clarity and precision, the dashboard I have personally created and refined emerges as a particularly useful, indispensable, and innovative tool. It provides an organized, transparent, and highly user-friendly framework that enables a deep and nuanced comprehension of these intricate interactions, breaking down complex data into an accessible and actionable format for all stakeholders involved. By offering critical, evidence-based insights derived from the analysis, the dashboard serves as a vital and transformative resource for policymakers, researchers, academics, healthcare professionals, and other key decision-makers who are tasked with shaping the future of global health and economic development. These insights have the potential to significantly influence and direct future policy decisions, fostering a more equitable, inclusive, and just approach to addressing global health challenges and promoting economic development. This tool not only illuminates the current state of affairs with clarity but also paves the way for informed, data-driven strategies that can help bridge the widening gap between resource-rich and resource-poor regions. By doing so, it ultimately contributes to the creation of a more balanced, prosperous, and healthier world, where the benefits of economic and health advancements can be shared more equitably across all nations and communities.

7. References

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